SIMI SHRIVASTAVA

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TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, scikit-learn, PyTorch, TensorFlow/Keras, SciPy, statsmodels, LightGBM, NLTK, gensim, Streamlit), SQL, R, MATLAB
- **Tools & Platforms:** Snowflake, ClickHouse, DuckDB, Spark, dbt, AWS (EC2, S3, ECR, ECS, Lambda, CloudWatch), Docker, Tableau, Power BI, Git, JIRA, Agile
- **Data Science:** Statistical Modeling, Predictive Modeling, A/B Testing, Causal Inference, Time-Series Forecasting, Hypothesis Testing, EDA, Feature Engineering, ETL Validation
- **Machine Learning:** Classification, Regression, NLP, Deep Learning (CNNs, Diffusion Models, RNNs), Model Evaluation (ROC/AUC, F1, WAPE), Cross-Validation, Hyperparameter Tuning

WORK EXPERIENCE

Data Science Intern | Canaria, Remote, US **Jun 2025 – Aug 2025**

- Built a salary benchmarking pipeline in Python and SQL on 20M+ records from 15+ job boards, cutting manual review time by 60%
- Designed anomaly detection (IQR, MAD) for multi-source salary data, cross-validated against BLS benchmarks to ensure consistency across 15+ source feeds
- Migrated the pipeline from pandas to Spark and ClickHouse, cutting end-to-end processing time by 70%

Data Analyst | Cognizant Technology Solutions (Client: LSEG), India **Nov 2019 – Jul 2024**

- Analyzed stock and index data in Python and SQL, finding trading patterns that improved trend prediction accuracy by 20%
- Created Tableau and Power BI dashboards tracking index performance and rebalancing, used by clients and senior leadership for market monitoring and portfolio decisions
- Developed an ML bug classifier (kNN, Decision Trees, word2vec) with JIRA integration, reducing validation from 1,500 to 4 minutes and saving $14K/mo
- Automated financial reconciliation during a platform migration using Macros and SQL, reducing an 8-hour process to under 5 minutes
- Delivered zero-downtime migration of a $15M financial database within a 48-hour window, coordinating across DBA, network, and dev teams

EDUCATION

University of Maryland, College Park, MD **Aug 2024 – May 2026**

Master of Science in Data Science | GPA: 3.86/4.0

Coursework: *Machine Learning, Deep Learning, Probability & Statistics, Big Data Systems, NLP, Cloud Computing, Algorithms for Data Science*

CSVTU, India **Jul 2015 – May 2019**

Bachelor of Engineering in Electronics and Telecommunication | GPA: 8.53/10.00

PROJECTS

Causal Demand Forecasting and Promotion ROI Analytics | *Python, LightGBM, Prophet, Snowflake, dbt, Streamlit*

- Compared four forecasting models on 3M+ grocery transactions with walk-forward validation (21 folds), best at 8.2% WAPE with LightGBM
- Measured +5% promotion lift with OLS counterfactual and quantified earthquake demand shocks from +21% panic buying to -11% supply disruption

Product Experimentation Analytics Framework | *Python, DuckDB, SQL, Tableau, Streamlit*

- Built an A/B testing framework on 541K transactions with SRM checks, a SQL metric layer in DuckDB, and segment testing, finding +3.85pp conversion lift (p=0.042) and +19.4% on mobile, delivered via Tableau and Streamlit

TabDDPM: Synthetic Patient Data for Readmission Prediction | *Python, PyTorch*

- Implemented TabDDPM in PyTorch on 101K hospital records (7.8:1 class imbalance), outperforming SMOTE and CTGAN with a 0.6616 ROC-AUC
- Improved minority-class F1 from ~0.05 to ~0.27 through threshold tuning, showing miscalibration mattered more than synthetic data quality